

EE558 - Optical Fiber Communication Systems

Prof. Alan Willner

List of Projects

Please check your first 4 choices for the project. If you wish, you can also choose a new topic (see last choice). Please return this sheet by 5 pm on Friday, Oct. 12 at 5 pm to Long Li (longl@usc.edu). If your choice is not received by Oct. 12 at 5 pm, you may be assigned a random topic.

(Note: Please choose 4 topics, since you probably won't get your first choice.)

Topics	1st Choice	2nd Choice	3rd Choice	4th Choice
1) Multiple-wavelength & tunable lasers				
2) Unconventional fibers (e.g., dispersion compensating, dispersion flattened, LEAF, NZ-DSF, and PM)				
3) Tunable filters and λ (de)multiplexers				
4) Erbium-doped fiber amplifiers				
5) Gain equalization and transients				
6) Nonlinear effects (XPM/FWM/SRS)				
7) Optical and electrical disp. compensators				
8) Polarization mode dispersion				
9) Forward-error-correction coding				
10) Spectrally-efficient modulation formats (PolMux, PSK, QPSK, QAM)				
11) Orthogonal Freq. Div. Muxing (OFDM)				
12) C-, L-, and S-band systems (wideband and Raman amplifiers)				
13) Optical switches (MEMS, etc.)	1			
14) High-speed time-division-multiplexing				4
15) Wavelength-division-multiplexing				
16) Heterodyne/Homodyne/Intradyne detection				

- 17) Free-space IR (infrared) comm.
- 18) Network architectures (OADM, OXC)
- 19) Network issues (contention resolution, control, protection, and restoration)
- 20) All-optical wavelength conversion
- 21) All-optical packet switching
- 22) Orthogonal-spatial-mode transmission
- 23) Multi-core fiber transmission systems
- 24) Optical performance monitoring
- 25) Subcarrier multiplexing (analog, CATV)
- 26) Microwave photonics
- 27) Data standards (SONET, Ethernet)
- 28) Fiber-to-the-home (Access & PONs)
- 29) What do you do with a Gb/s? (services)
- 30) Optical interconnections – On-chip
- 31) Optical Comm in Data Centers
- 32) Optical quantum communications
- 33) Personal Write-In Choice: (give topic)

3

2